Code-A

Time : 3:00 Hrs. | M. Marks : 720

Pre NEET Mock Test-4

(*for* NEET-2021)

(Complete Syllabus of Class XI & XII)

Instructions :

(i) There are two sections in each subject, i.e. Section-A & Section-B. You have to attempt all 35 questions from Section-A & only 10 questions from Section-B out of 15.

(ii) Each question carries 4 marks. For every wrong response 1 mark shall be deducted from the total score. Unanswered /unattempted questions will be given no marks.

(iii) Use blue/black ballpoint pen only to darken the appropriate circle.

(iv) Mark should be dark and completely fill the circle.

(v) Dark only one circle for each entry.

(vi) Dark the circle in the space provided only.

(vii) Rough work must not be done on the Answer sheet and do not use **white-fluid** or any other **rubbing material** on the Answer sheet.

PHYSICS

Choose the correct answer :

SECTION-A

1. A telescope has an objective lens of 20 cm diameter and is situated at a distance of 2 km from two objects. The minimum distance between the two objects, which can be resolved by the telescope, when mean wavelength of light is 6000 Å, is order of

(1) 7 m (2) 7 cm

(3) 7 mm (4) 7 km

2. When a Uranium isotope ${}^{235}_{92}U$ is bombarded with a thermal neutron, it generates ${}^{141}_{56}Ba$, three neutrons and :

(1) ${}^{101}_{36}Kr$

(2) ${}^{91}_{40}Zr$

(3) ${}^{92}_{36}Kr$

(4) ${}^{144}_{56}Ba$

3. A short electric dipole has dipole moment of $3\sqrt{2}\ \ 10^{-10}$ C m. The electric potential due to the dipole at a point which is at a distance 0.3 m from the centre of the dipole, situated on a line making an angle of 45° with the dipole axis is : $\left(\frac{1}{4\pi\varepsilon_0} = 9\times10^9\ \text{N m}^2/\text{C}^2\right)$

(1) 300 V (2) 30 V

(3) 200 V (4) 20 V

4. A ray is incident at an angle of incidence *i* on one surface of a small angled prism (with angle of prism 8°) and emerges normally from the opposite surface. If the refractive index of the material of prism is $\frac{3}{2}$, then the angle of incidence nearly equal to

(1) 12° (2) 8°

(3) 14° (4) 6°

5. A body weighs 80 N on the surface of the earth. What is its weight at a height equal to radius of the earth?

(1) 40 N (2) 60 N

(3) 30 N (4) 20 N

6. The angular momentum of an electron in hydrogen atom for $n = 4$, is

(1) $\frac{h}{2\pi}$ (2) $\frac{3h}{2\pi}$

(3) $\frac{h}{\pi}$ (4) $\frac{2h}{\pi}$

7. A capillary tube of radius r and length $3h$ is immersed in water. The water rises upto a height h. The corresponding mass is m. What mass of water will rise in the capillary if the radius of the tube is halved?

(1) m (2) $\frac{m}{2}$

(3) $2m$ (4) $4m$

8. The least count of a vernier caliper is 0.01 cm and 10 divisions of the vernier scale is equal to n divisions of main scale. The value of n will be (1 division of main scale is 1 mm)

(1) 6 (2) 9

(3) 10 (4) 12

9. The magnetic susceptibility of a medium is –0.08, what will be its relative permeability?

(1) 1.8 (2) 1.08

(3) 0.08 (4) 0.92

10. Which among the following quantities, is always negative for a particle executing S.H.M.? (Symbols have their usual meaning)

(1) $\vec{r}\cdot\vec{v}$ (2) $\vec{v}\cdot\vec{a}$

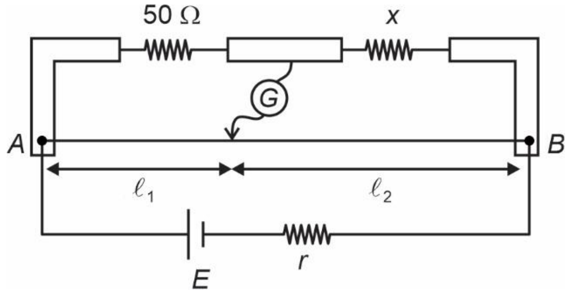
(3) $\vec{a}\cdot\vec{r}$ (4) $\vec{r}\times\vec{v}$

11. The value of energy equivalent of 4 atomic mass unit (a.m.u) in MeV will be

(1) 3726 MeV (2) 2762 MeV

(3) 3264 MeV (4) 2332 MeV

12. In a practical meter bridge circuit as shown, when one more resistance of 50 Ω is connected parallel with unknown resistance 'x', then ratio $\frac{\ell_1}{\ell_2}$ becomes '2'. The value of x is (ℓ_1 is balance length, AB is uniform wire)



(1) 100 Ω (2) 50 Ω

(3) 200 Ω (4) 25 Ω

13. The average rotational kinetic energy for the diatomic gas molecules is (k_B is Boltzmann constant and T is absolute temperature)

(1) $\frac{3}{2}k_BT$ (2) $\frac{7}{2}k_BT$

(3) k_BT (4) $\frac{5}{2}k_BT$

14. In an electromagnetic wave, the amplitudes of magnetic field (B_0) and electric field (E_0) in free space are related as :

(1) $B_0 = E_0$ (2) $B_0 = \frac{E_0}{c^2}$

(3) $B_0 = E_0\sqrt{\frac{\varepsilon_0}{\mu_0}}$ (4) $B_0 = E_0\sqrt{\mu_0\varepsilon_0}$

15. The dimensional formula of velocity gradient is

(1) $[M^0L^2T^{-1}]$ (2) $[M^0L^0T^{-1}]$

(3) $[ML^0T^{-1}]$ (4) $[MLT^{-1}]$

16. A long solenoid of 100 cm length having 100 turns, carries a current of 5 A. The magnetic field at the end of the solenoid is : ($\mu_0 = 4\pi \times 10^{-7}$ T m A^{-1}]

(1) 3.14×10^{-4} T (2) 3.14×10^{-6} T

(3) 6.28×10^{-4} T (4) 6.28×10^{-6} T

17. Taking into account of the significant figures, what is the value of 23.623 cm + 8.7 cm?

(1) 32.3 (2) 32.323

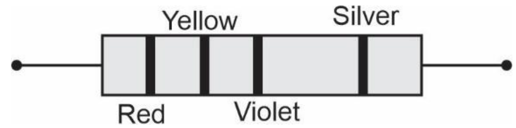
(3) 32.32 (4) 32

18. Light of frequency 3 times the threshold frequency is incident on a photosensitive material. What will be the saturation photoelectric current, if the intensity is halved keeping frequency constant?

(1) Doubled (2) Halved

(3) One fourth (4) Zero

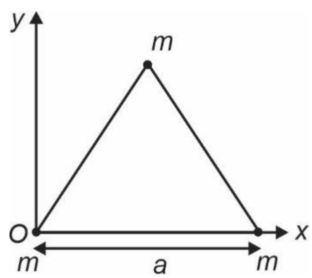
19. The color code of a carbon resistor is given below



The resistance of the resistor will be

(1) $24 \times 10^7\ \Omega \pm 10\%$
(2) $25 \times 10^4\ \Omega \pm 10\%$
(3) $35 \times 10^6\ \Omega \pm 5\%$
(4) $27 \times 10^5\ \Omega \pm 5\%$

20. Three particles each of mass *m* are placed at the corners of an equilateral triangle of side *a* as shown in the figure. The *X*-co-ordinate of centre of mass will be



(1) $\frac{\sqrt{3}a}{2}$
(2) $\frac{3a}{4}$
(3) $\frac{a}{2}$
(4) a

21. In a bipolar junction transistor (BJT)

(1) the base has the least concentration of impurity.
(2) the emitter has least concentration of impurity
(3) all the three regions have equal concentration of impurity
(4) the collector has least concentration of impurity

22. The mean free path for gas molecules with molecular diameter *d* and number density *n* is proportional to

(1) $\frac{1}{d}$
(2) d^2
(3) $\frac{1}{n}$
(4) $\frac{1}{n^2}$

23. If the density of a helium gas is 0.1 kg/m^3 at pressure 249 kPa, then temperature of the gas will be nearly $\left[R = \frac{25}{3}\text{J mol}^{-1}\text{ K}^{-1}\right]$

(1) 1195 °C
(2) 922 °C
(3) 27 °C
(4) 327 °C

24. An electric current of 8 A exists in a metal wire of cross section 10^{-4} m^2 and length 2 m. The drift speed of the free electrons in the wire will be (Given $n = 5 \times 10^{28}$ m^{-3})

(1) 2×10^{-3} m/s
(2) 1×10^{-3} m/s
(3) 2×10^{-2} m/s
(4) 1×10^{-5} m/s

25. A ball is thrown vertically downwards with speed 30 m/s from the top of the tower of height 80 m. The speed of the ball with which it hits the grounds is (g = 10 m/s^2)

(1) 40 m/s
(2) 60 m/s
(3) 50 m/s
(4) 30 m/s

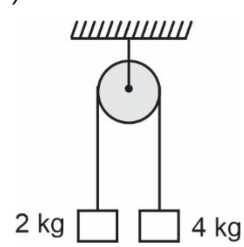
26. A light wire of length *L*, area of cross section *A* is hanging from fixed support. If a mass *M* is suspended from its free end, then strain produced in the wire will be (symbols have their usual meaning)

(1) $\frac{Mg}{2YA}$
(2) $\frac{Mg}{YA}$
(3) $\frac{2Mg}{YA}$
(4) Zero

27. A fork *P* of unknown frequency gives six beats per second when sounded with another fork *Q* of frequency 260 Hz. The fork *P* is now loaded with a piece of wax and again six beats per second is heard, then frequency of fork *P* is

(1) 260 Hz
(2) 254 Hz
(3) 266 Hz
(4) 250 Hz

28. Two bodies of masses 2 kg and 4 kg are tied to the ends of a massless string. The string passes over a pulley which is frictionless as shown in figure. If the system is released from rest, then tension in the string connecting the bodies will be (g = 10 m/s^2)



(1) 80 N
(2) $\frac{80}{3}$ N
(3) 20 N
(4) $\frac{20}{3}$ N

29. On removing dielectric slab from an isolated charged capacitor, the quantity that remains constant is

(1) Capacitance (2) Charge

(3) Potential difference (4) Energy stored

30. In Young's double slit experiment using yellow and blue lights of wavelengths 560 nm and 420 nm respectively, the value of n for which n^{th} order of yellow bright fringe coincides with $(n + 1)^{th}$ order of blue bright fringe is

(1) 3 (2) 4

(3) 2 (4) 5

31. The sun delivers 10^5 W/m^2 electromagnetic flux to the earth's surface. The total power that is incident on the roof of dimensions (20×20) m^2 will be

(1) 2×10^7 W (2) 10^6 W

(3) 2×10^6 W (4) 4×10^7 W

32. The resistivity (ρ) of a metallic conductor on increasing temperature (T)

(1) Will increase

(2) Will decrease

(3) Remains constant

(4) May decrease or remain constant

33. The quantities of heat required to raise the temperature of two metallic spheres of same mass having specific heat capacity S_1 and S_2 ($3S_1 = 4S_2$) through 1 K are in the ratio:

(1) $\frac{4}{3}$ (2) $\frac{9}{16}$

(3) $\frac{1}{3}$ (4) $\frac{2}{3}$

34. A force $\vec{F} = (\hat{i} + 2\hat{j})$ N acts at O, the origin of the co-ordinate system. The torque about the point (1, –1, 1) m is

(1) $(2\hat{i} - \hat{j} - 3\hat{k})$ N m (2) $(\hat{i} - \hat{j} - \hat{k})$ N m

(3) $(2\hat{i} + 2\hat{j} - \hat{k})$ N m (4) $(-\hat{i} + \hat{j} + 2\hat{k})$ N m

35. The electric potential at any point (x, y, z) is given as $V = (2y^3)$ volt. The electric field at point (2, 1, 0) is (x, y, z all are in metre)

(1) $6\hat{j}$ V m^{-1} (2) $-6\hat{j}$ V m^{-1}

(3) $2\hat{i}$ V m^{-1} (4) $-2\hat{i}$ V m^{-1}

SECTION-B

36. The light is incident from air to a medium of the refractive index $\frac{4}{3}$. The Brewster's angle for this scenario will be

(1) 37° (2) 45°

(3) 53° (4) 90°

37. For which of the following conditions, the width of the depletion region in a p-n junction diode decreases

(1) Reverse bias

(2) Forward bias

(3) Both forward bias and reverse bias

(4) Decreasing the doping concentration

38. A charge q is uniformly distributed on a spherical shell of radius R. The electric field at a distance $\frac{4R}{3}$ from the centre will be

(1) $\frac{9q}{64\pi\varepsilon_0 R^2}$ (2) $\frac{q}{4\pi\varepsilon_0 R^3}$

(3) $\frac{3q}{16\pi\varepsilon_0 R^2}$ (4) Zero

39. The value of 64 J energy is equal to

(1) 2×10^{20} eV (2) 10^{19} eV

(3) 4×10^{20} eV (4) 2×10^{19} eV

40. An alternating voltage E (in volt) = $300\sqrt{2}\sin(100t)$ is connected to a 2 μF capacitor through an AC ammeter. The reading of the ammeter would be

(1) 30 mA (2) 60 mA

(3) 20 mA (4) 40 mA

41. The de Broglie wavelength of an electron having 320 eV of energy is nearly (1 eV = 1.6×10^{-19} J, mass of electron = 9.1×10^{-31} kg, planck's constant = 6.6×10^{-34} J.s)

(1) 70 Å (2) 0.7 Å

(3) 1.4 Å (4) 14 Å

42. In adiabatic expansion of an ideal gas

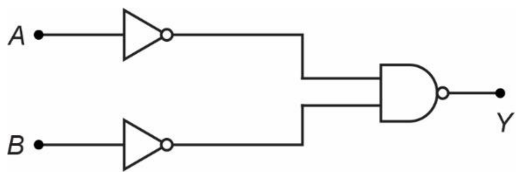
(1) Heat is gained or lost by the gas

(2) Heat is neither gained nor lost by the gas

(3) Temperature remains constant

(4) Volume remains constant

43. The following figure shows the combination of logic gates.



This combination is equivalent to

(1) NAND gate

(2) OR gate

(3) NOR gate

(4) XOR gate

44. Resistance of which of the following solids decreases on increasing temperature? (Consider no change in dimensions)

(1) Metals

(2) Semiconductors only

(3) Insulators only

(4) Insulator and semiconductors

45. An alternating potential $V = V_0 \sin\omega t$ is applied across a circuit. As a result the current $I = I_0 \sin\left(\omega t - \frac{\pi}{3}\right)$ flows through it. The average power consumed in the circuit is

(1) 0.5 V_0I_0 (2) 0.707 V_0I_0

(3) 0.25 V_0I_0 (4) 1.414 V_0I_0

46. What is the radius of curvature of an equiconvex lens made of glass of refractive index $\frac{5}{3}$, if its focal length is 40 cm?

(1) 10 cm (2) $\frac{160}{3}$ cm

(3) 20 cm (4) $\frac{140}{3}$ cm

47. The maximum kinetic energy of emitted electrons in a photoelectric effect does not depend upon

(1) Frequency (2) Wavelength

(3) Work function (4) Intensity

48. The motion of a particle along a straight line is described by equation $x = 12t - t^3$ where x is in meter and t is in second. The time instant when velocity of the particle becomes zero is

(1) 2 s (2) 1 s

(3) 4 s (4) 3 s

49. A block of mass 4 kg rests on a rough horizontal surface. If a horizontal force of 12 N is applied on the block, then the frictional force by the surface on the block is

($\mu_k = 0.4$, $\mu_s = 0.5$ and $g = 10$ m/s^2)

(1) 16 N (2) 12 N

(3) 20 N (4) 8 N

50. The radius of gyration of a spherical shell of mass M and radius R about its tangential axis will be

(1) $\sqrt{\frac{2}{5}}R$ (2) $\sqrt{\frac{7}{5}}R$

(3) $\sqrt{\frac{2}{3}}R$ (4) $\sqrt{\frac{5}{3}}R$

CHEMISTRY

SECTION-A

51. Manufacture of SO_3 is given by the following reaction

$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$

If the equilibrium constant K_c is 1.7×10^2 at 300K, then the value of $\Delta_rG°$ at the same temperature will be

(1) 8.314 J mol^{-1} K^{-1} × 300 K × ln (3.4×10^2)

(2) –8.314 J mol^{-1} K^{-1} × 300 K × ln (1.7×10^2)

(3) –8.314 J mol^{-1} K^{-1} × 300 K × ln (2.7×10^2)

(4) 8.314 J mol^{-1} K^{-1} × 300 K × ln (2.7×10^2)

52. Which of the following has maximum number of atoms?

(1) 10 g of N_2(g) [Atomic mass of N = 14u]

(2) 10 g of O_2(g) [Atomic mass of O = 16u]

(3) 10 g of O_3(g) [Atomic mass of O = 16u]

(4) 10 g of Cu(s) [Atomic mass of Cu = 63.5u]

53. Which of the following is not correct about carbon monoxide?

(1) Water soluble gas

(2) It is a powerful reducing agent

(3) Form complex with haemoglobin

(4) Highly poisonous in nature

54. The calculated spin only magnetic moment of Co^{2+} ion is

(1) 4.90 BM (2) 5.92 BM

(3) 2.84 BM (4) 3.87 BM

55. Which of the following is a natural polymer?

(1) Cellulose

(2) Nylon 6, 6

(3) Neoprene

(4) Buna-S

56. Which of the following is a basic amino acid?

(1) Arginine (2) Serine

(3) Cysteine (4) Threonine

57. A mixture of H_2 and O_2 gases in a cylinder contains 10 g H_2 and 64 g of O_2. If the total pressure of the mixture of the gases in the cylinder is 35 bar, the partial pressure of H_2 gas is

(1) 2 bar (2) 5 bar

(3) 25 bar (4) 20 bar

58. Which of the following is a type of partition chromatography?

(1) Thin layer chromatography

(2) Paper chromatography

(3) Column chromatography

(4) Adsorption chromatography

59. For the reaction $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$, the correct option is

(1) $\Delta_r H > 0$ and $\Delta_r S > 0$

(2) $\Delta_r H > 0$ and $\Delta_r S < 0$

(3) $\Delta_r H < 0$ and $\Delta_r S < 0$

(4) $\Delta_r H < 0$ and $\Delta_r S > 0$

60. Urea reacts with water to form A which will decompose to form B. B when passed through Co^{3+}(aq), yellow coloured solution C is formed. What is the formula of C from the following?

(1) $[Co(NH_3)_6]^{3+}$ (2) CoO

(3) $Co_2(SO_4)_3$ (4) Co_3O_4

61. On electrolysis of aqueous NaCl solution using platinum (Pt) electrodes, the product obtained at anode will be

(1) Hydrogen gas

(2) Oxygen gas

(3) Chlorine gas

(4) Ozone gas

62. An element has a body centered cubic (bcc) structure with a cell edge length of 288 pm. The shortest interatomic distance is

(1) $\frac{4}{\sqrt{2}} \times 288$ pm (2) $\frac{\sqrt{2}}{4} \times 288$ pm

(3) $\frac{\sqrt{3}}{2} \times 288$ pm (4) $\sqrt{3} \times 288$ pm

63. Identify the incorrect statement

(1) Inhibitors reduces the rate of reaction

(2) Catalyst reduces the activation energy

(3) Rate constant increases on increasing temperature

(4) Catalyst catalyses the non-spontaneous reaction

64. Which of the following alkane cannot be made in good yield by Wurtz reaction?

(1) n-Butane

(2) n-Hexane

(3) 3, 4-Dimethylhexane

(4) 3-Methylhexane

65. Identify the incorrect statement

(1) Interstitial compounds are usually non-stoichiometric and are neither typically ionic nor covalent

(2) The oxidation state of manganese in manganate and permanganate ions are same

(3) Oxidising power of VO_2^+ is lesser than that of $Cr_2O_7^{2-}$

(4) Zn and Cd are not regarded as transition metals

66. Which of the following is a cationic detergent?

(1) Cetyltrimethyl ammonium bromide

(2) Sodium laurylsulphate

(3) Sodium dodecylbenzenesulphonate

(4) Stearic acid

67. A benzyl carbocation is more stable than a secondary butyl carbocation mainly because of which of the following?

(1) –I effect of phenyl group

(2) Hyperconjugation

(3) +I effect of phenyl group

(4) Resonance effect of phenyl group

68. Which of the following is the correct order of increasing field strength of ligands to form coordination compounds?
(1) $CO < C\bar{N} < en < NH_3$
(2) $C\bar{N} < CO < NH_3 < en$
(3) $NH_3 < en < C\bar{N} < CO$
(4) $NH_3 < en < CO < C\bar{N}$

69. Identify the correct statement from the following
(1) Vapour phase refining is carried out for Titanium by Mond process
(2) Pig iron is the purest form of iron
(3) Blister copper has blistered appearance due to evolution of SO_2
(4) Pig iron is prepared from cast iron by oxidising impurities

70. Which of the following sets of molecules will have zero dipole moments?
(1) Boron trifluoride, carbon tetrachloride, 1, 4-dichlorobenzene, sulphur dioxide
(2) Ammonia, carbon dioxide, 1, 3-dichlorobenzene, water
(3) Boron trifluoride, 1,3-dichlorobenzene, ammonia, ethanal
(4) Boron trifluoride, 1,4-dichlorobenzene, carbon dioxide, carbon tetrachloride

71. Match the following and identify the correct option

(a)	$CO(g) + N_2(g)$	(i)	An electron precise hydride
(b)	Permanent hardness of water	(ii)	Producer gas
(c)	CH_4	(iii)	Peroxide linkage
(d)	H_2O_2	(iv)	$MgCl_2 + MgSO_4$

	(a)	**(b)**	**(c)**	**(d)**
(1)	(i)	(iv)	(ii)	(iii)
(2)	(ii)	(iv)	(i)	(iii)
(3)	(iii)	(ii)	(iv)	(i)
(4)	(ii)	(i)	(iv)	(iii)

72. The number of protons, neutrons and electrons in $^{140}_{58}Ce$ respectively are
(1) 140, 58 and 58 (2) 58, 58 and 82
(3) 58, 82 and 58 (4) 82, 58 and 58

73. The number of Faradays (F) required to produce 9 g of aluminium from molten $AlCl_3$ is (Atomic mass of Al = 27 g mol^{-1})
(1) 2 (2) 3
(3) 1 (4) 4

74. Elimination reaction of 1-Bromohexane to form hex-1-ene is
(a) Dehydration reaction
(b) Dehydrohalogenation reaction
(c) β-Elimination reaction
(1) (a) and (b) only (2) (b) and (c) only
(3) (a) and (c) only (4) (a), (b) and (c)

75. Identify the correct statements from the following
(a) Silicones are used in surgical and cosmetic plants
(b) $CO_2(g)$ causes greenhouse effect
(c) Quartz is extensively used as piezoelectric material
(d) Carbon atoms in fullerenes are sp^2 and sp^3 hybridised
(1) (a) and (c) only (2) (b) and (c) only
(3) (c) and (d) only (4) (a), (b) and (c) only

76. Identify the incorrect match

	Name		IUPAC Official Name
a.	Unnilseptium	(i)	Bohrium
b.	Unnilpentium	(ii)	Dubnium
c.	Unnilquadium	(iii)	Rutherfordium
d.	Ununbium	(iv)	Nobelium

(1) b, (ii) (2) c, (iii)
(3) a, (i) (4) d, (iv)

77. The freezing point depression constant (K_f) of benzene is 5.12 K kg mol^{-1}. The freezing point depression for the solution of molality 0.01 m containing a non-electrolyte solute in benzene is (rounded off upto two decimal places)
(1) 0.1 K (2) 0.2 K
(3) 0.05 K (4) 0.5 K

78. What is the change in oxidation number of carbon in the following reaction?
$N_2O_4(g) + 3CO(g) \rightarrow N_2O(g) + 3CO_2(g)$
(1) +2 to +4 (2) 0 to +4
(3) 0 to +2 (4) +4 to –4

79. The rate constant for a first order reaction is 2.303×10^{-2} s^{-1}. The time required to reduce 8.0 g of the reactant to 1.0 g is

(1) 60 s (2) 90 s

(3) 120 s (4) 150 s

80. The mixture which shows negative deviation from Raoult's law is

(1) n-hexane + n-heptane

(2) Acetone + chloroform

(3) Ethanol + Acetone

(4) Acetone + carbon disulphide

81. Which of the following oxoacids of sulphur has –O–O– linkage?

(1) H_2SO_5 (2) $H_2S_2O_4$

(3) $H_2S_2O_7$ (4) $H_2S_2O_3$

82. Most effective ion for the coagulation of As_2S_3 sol is

(1) Al^{3+} (2) Mg^{2+}

(3) Cl^- (4) SO_4^{2-}

83. Lactose on hydrolysis gives

(1) α-D-glucose and α-D-fructose

(2) β-D-glucose and β-D-galactose

(3) β-D-glucose and β-D-glucose

(4) α-D-glucose and β-D-galactose

84. Find out the solubility of $Mg(OH)_2$ in 0.1 M NaOH. Given that solubility product of $Mg(OH)_2$ is 1.8×10^{-11}.

(1) 3.24×10^{-10} M (2) 1.8×10^{-9} M

(3) 3.24×10^{-9} M (4) 1.8×10^{-10} M

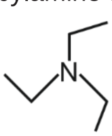
85. The incorrect option for free expansion of ideal gas under isothermal condition is

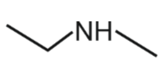
(1) $\Delta H = 0$ (2) $\Delta S = 0$

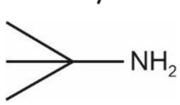
(3) $\Delta T = 0$ (4) w = 0

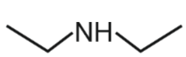
SECTION-B

86. Which of the following amines will give carbylamine test?

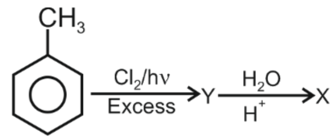
(1) 

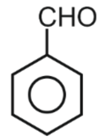
(2) 

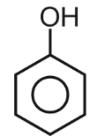
(3) 

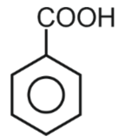
(4) 

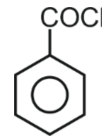
87. Identify compound X in the following sequence of reactions



(1) 

(2) 

(3) 

(4) 

88. Reaction between methanal and ethanal in presence of dilute NaOH is known as

(1) Cannizzaro's reaction

(2) Cross Cannizzaro's reaction

(3) Cross aldol condensation

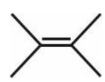
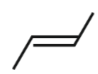
(4) Aldol condensation

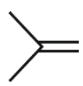
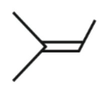
89. Identify a molecule which does not exist

(1) Be_2 (2) N_2

(3) B_2 (4) C_2

90. An alkene on ozonolysis gives methanal as one of the product. Its structure is

(1)  (2) 

(3)  (4) 

91. Which of the following metal ions found primarily on the outside of cells, being located in blood plasma and in the interstitial fluid which surrounds the cells?

(1) Copper

(2) Iron

(3) Sodium

(4) Magnesium

92. Cleavage of $CH_2 = CH - O - CH_2 - CH_3$ with HI gives

(1) $CH_2 = CH - I + CH_3 - CH_2 - OH$

(2) $CH_2 = CH - I + CH_3OH$

(3) $CH_3 - \overset{\overset{\large O}{||}}{C} - H + CH_3 - CH_2 - I$

(4) $CH_2 = CH - OH + CH_3I$

93. Match the following

	Oxide		Nature
(a)	N_2O	(i)	Basic
(b)	SO_3	(ii)	Neutral
(c)	Na_2O	(iii)	Acidic
(d)	PbO	(iv)	Amphoteric

Which of the following is correct option?

	(a)	**(b)**	**(c)**	**(d)**
(1)	(ii)	(iii)	(i)	(iv)
(2)	(i)	(ii)	(iv)	(iii)
(3)	(iii)	(i)	(iv)	(ii)
(4)	(iv)	(ii)	(i)	(iii)

94. For the formation of NaOH using Castner Kellner cell, cathode used is of metal

(1) Hg (2) Pt

(3) Cu (4) Na

95. Reaction between carbon dioxide and methyl magnesium bromide followed by hydrolysis will give

(1) CH_3CHO

(2) CH_3CH_2OH

(3) CH_3COOH

(4) CH_3CH_3

96 Lewis acid among the following is

(1) BF_3 (2) NH_3

(3) CH_3OCH_3 (4) H_2O

97. Which of the following sets of quantum numbers is not possible?

(1) n = 4, l = 3 , m = –2, $s=+\frac{1}{2}$

(2) n = 3, l = 0, m = –1, $s=+\frac{1}{2}$

(3) n = 4, l = 0, m = 0, $s=-\frac{1}{2}$

(4) n = 3, l = 2, m = –2, $s=+\frac{1}{2}$

98. Minimum ionic radii among the following is of

(1) Ce^{3+}

(2) Pr^{3+}

(3) Sm^{3+}

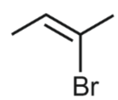
(4) Eu^{3+}

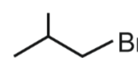
99. Ambidentate ligand is

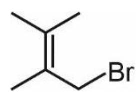
(1) $C_2O_4^{2-}$ (2) NO_2^-

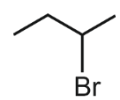
(3) $EDTA^{4-}$ (4) H_2O

100. Fastest rate of S_N1 reaction is given by the which molecule?

(1) 

(2) 

(3) 

(4) 

BOTANY

SECTION-A

101. Select the **incorrect** about *Equisetum*.

(1) Has vascular tissues

(2) Homosporous

(3) Has strobili or cones

(4) Produce two types of spores

102. Select the **correct** match.

(1) Phenylketonuria – Autosomal recessive disorder

(2) Thalassemia – Qualitative defect in synthesis of globin

(3) Colour blindness – Y-linked

(4) Myotonic dystrophy – Autosomal recessive disorder

103. Crossing over or recombination takes place in

(1) Zygotene (2) Diplotene

(3) Leptotene (4) Pachytene

104. Which of the following is not an attribute of a population?

(1) Birth rate

(2) Death rate

(3) Sex ratio

(4) Sex of an individual

105. Identify the **incorrect** statement.
   (1) Sapwood is peripheral part of wood
   (2) Heartwood has tannins, resins, oils etc
   (3) Heartwood provides mechanical support
   (4) Sapwood lacks dead vessels and tracheids

106. Snow blindness refers to
   (1) Various type of skin cancers
   (2) Aging of skin
   (3) Inflammation of cornea due to UV-B
   (4) Cataract of eye

107. In the equation $\frac{dN}{dt} = rN\ \frac{K-N}{K}$, K refers to
   (1) Population density
   (2) Intrinsic rate
   (3) Carrying capacity
   (4) The base of natural logarithms

108. Which of the following statements is **incorrect** about DNA?
   (1) Guanine is bonded with Cytosine with three H-bonds.
   (2) The two chains are coiled in a right handed fashion.
   (3) Backbone is constituted by sugar-phosphate.
   (4) A purine comes opposite to a purine only.

109. Some dividing cells exit the cell cycle and enter vegetative inactive stage called
   (1) G_1 (2) G_2
   (3) G_0 (4) S

110. Flower is perigynous in
   (1) Mustard
   (2) Brinjal
   (3) Potato
   (4) Peach

111. An aquatic plant which requires biotic pollinating agent is
   (1) Water Lily
   (2) *Hydrilla*
   (3) *Zostera*
   (4) *Vallisneria*

112. Which of the following enzyme of transcription helps in opening of DNA helix as well as transcribing it?
   (1) DNA polymerase (2) RNA polymerase
   (3) DNA helicase (4) Topoisomerase

113. Select the **incorrect** statement.
   (1) Formation of peptide bond requires energy
   (2) Charging of tRNA is called aminoacylation of tRNA.
   (3) Ribosome in its inactive state exists as two subunits.
   (4) First phase of translation is binding of mRNA to ribosomes.

114. Which of the following does **not** happen in G_2 phase?
   (1) Cell organelles duplication
   (2) Tubulin synthesis
   (3) Replication of DNA
   (4) RNA synthesis.

115. Inferior ovary is seen in epigynous flowers as in
   (1) Guava
   (2) Plum
   (3) Rose
   (4) Pea

116. Agar is obtained from red algae such as
   (1) *Chlorella* and *Spirulina*
   (2) *Gelidium* and *Gracilaria*
   (3) *Laminaria* and *Sargassum*
   (4) *Anabaena* and *Volvox*

117. The plant parts which are haploid and present inside diploid part are
   (a) Embryo within the seed
   (b) Pollen grains within the Pollen sac
   (c) Seed inside the fruit
   (d) Female gametophyte within the ovule
   (e) Male gametes within the pollen grains.
   (1) (a), (d) and (e)
   (2) (b) and (c)
   (3) (b) and (d)
   (4) (c), (d) and (e)

118. Select the **incorrect** statement.
   (1) Montreal Protocol was signed to control emission of greenhouse gases.
   (2) There is good ozone in stratosphere.
   (3) The thickness of ozone is measured in Dobson unit.
   (4) CFCs does not harm Ozone layer.

119. If the length of *E. coli* DNA is 1.36 mm, then what will be the number of base pairs if distance between two consecutive base pairs is 0.34 nm?

(1) 6.6×10^9 bp

(2) 3.3×10^9 bp

(3) 4.0×10^6 bp

(4) 2.3×10^6 bp

120. Nitrogenase is enzyme which converts N_2 into NH_3, it is

(1) Inhibited by pigment leghaemoglobin

(2) Oxygen sensitive

(3) Found in all prokaryotes

(4) Copper containing enzyme

121. Among the animals, which is the most species rich taxonomic group?

(1) Molluscs

(2) Insects

(3) Fishes

(4) Reptiles

122. Which of the following has ribosomes for protein synthesis?

(1) Peroxisomes

(2) Golgi bodies

(3) Lysosomes

(4) Rough endoplasmic reticulum

123. There were seven contrasting traits taken by Mendel for his experiments on pea. Which of the following was not taken?

(1) Flower colour (2) Stem height

(3) Pod colour (4) Pod weight

124. Match the following columns and select the **correct** option.

	Column I		Column II
(a)	*Clostridium butylicum*	(i)	Citric acid
(b)	*Aspergillus niger*	(ii)	Acetic acid
(c)	*Acetobacter aceti*	(iii)	Butyric acid
(d)	*Lactobacillus*	(iv)	Lactic acid

	(a)	(b)	(c)	(d)
(1)	(iv)	(iii)	(i)	(ii)
(2)	(ii)	(i)	(iii)	(iv)
(3)	(iii)	(i)	(ii)	(iv)
(4)	(iii)	(i)	(iv)	(ii)

125. Match the following columns w.r.t. mitosis and select the correct option.

	Column-I		Column-II
(a)	Prophase	(i)	Re-appearance of nuclear membrane
(b)	Metaphase	(ii)	Condensation of chromatin
(c)	Anaphase	(iii)	Separation of chromatids
(d)	Telophase	(iv)	Chromosomes align on equator

	(a)	(b)	(c)	(d)
(1)	(iii)	(ii)	(i)	(iv)
(2)	(ii)	(iv)	(iii)	(i)
(3)	(ii)	(iv)	(i)	(iii)
(4)	(iv)	(i)	(iii)	(ii)

126. Which of the following are **not** membrane bound?

(1) Inclusion bodies of prokaryotes

(2) Lysosomes

(3) Plant vacuoles

(4) Endoplasmic reticulum

127. Beetle with red and black markings – the ladybird and dragonflies are useful to get rid of

(1) Aphids and mosquitoes

(2) All fungal diseases

(3) Plant nematodes

(4) Insect predators

128. Match the trophic levels with their correct species names in grassland ecosystem and select the **correct** option.

	Trophic levels		Species
(a)	Tertiary Consumer	(i)	Grass
(b)	Primary Consumer	(ii)	Crow
(c)	Producers	(iii)	Vulture
(d)	Secondary Consumer	(iv)	Rabbit

	(a)	(b)	(c)	(d)
(1)	(iv)	(i)	(ii)	(iii)
(2)	(ii)	(iii)	(iv)	(i)
(3)	(iii)	(iv)	(i)	(ii)
(4)	(iii)	(iv)	(ii)	(i)

129. Name the plant growth regulator which helps in cell division and counteracts apical dominance?
(1) Gibberellin (2) Ethylene
(3) ABA (4) Cytokinin

130. Which of the following cause fruits like apple to elongate and improve its shape?
(1) Cytokinins (2) Gibberellins
(3) Auxin (4) Ethylene

131. Primary productivity depends on all of the given factors, **except**
(1) Plant species inhabiting a particular area.
(2) A variety of environmental factors.
(3) Photosynthetic capacity of plants.
(4) Various species of primary consumers.

132. Match the following essential elements and their functions in plants.

	Elements		**Functions**
(a)	Iron	(i)	Photolysis of water.
(b)	Magnesium	(ii)	Required for chlorophyll biosynthesis.
(c)	Chlorine	(iii)	Maintains anion cation balance
(d)	Manganese	(iv)	Helps to maintain ribosome structure

Select the **correct** option.

	(a)	**(b)**	**(c)**	**(d)**
(1)	(ii)	(iv)	(i)	(iii)
(2)	(iv)	(ii)	(iii)	(i)
(3)	(ii)	(iv)	(iii)	(i)
(4)	(iv)	(ii)	(i)	(iii)

133. Viroids have
(1) RNA
(2) DNA
(3) Protein Coat
(4) Proteinaceous envelope

134. India has how many biosphere reserves?
(1) 90 (2) 27
(3) 14 (4) 448

135. Ovule has integuments which encircle the nucellus except at the tip where a small opening is present called
(1) Chalaza (2) Hilum
(3) Micropyle (4) Funicle

SECTION-B

136. Fibrous roots originate from
(1) Radicle (2) Primary root
(3) Base of the stem (4) Lateral roots

137. Substrate level phosphorylation is
(1) NADH synthesis
(2) Direct ATP synthesis
(3) $NADPH_2$ synthesis
(4) $FADH_2$ synthesis

138. Father of Genetics is
(1) Sutton (2) Boveri
(3) Morgan (4) Mendel

139. Plasmolysis occurs when a plant cell is placed
(1) In a solution that is isotonic to the protoplasm
(2) In a solution that is hypertonic to the protoplasm
(3) In a solution that is hypotonic to the protoplasm
(4) In pure water

140. Select the **incorrect** statement about RuBisCO.
(1) It is the most abundant enzyme.
(2) It has much greater affinity for CO_2 than for O_2.
(3) In C_3 plants some O_2 does bind to RuBisCO.
(4) RuBP combines with O_2 to form 3 molecules of 3-PGA, this reaction is also catalysed by RuBisCO.

141. How many genotypes and phenotypes are there in ABO blood groups in humans?

	Genotype	**Phenotype**
(1)	6	4
(2)	4	6
(3)	6	6
(4)	4	4

142. Primary sludge contains
(1) Methanogens and fungi
(2) Bacterial flocs
(3) Soil and small pebbles
(4) Aerobic bacteria and anaerobic bacteria

143. Laminarin is stored food material in
(1) Red algae (2) Brown algae
(3) Green algae (4) Blue green algae

144. The transverse section of a plant part shows following anatomical features.

(a) Mesophyll not differentiated

(b) Conjoint vascular bundles

(c) Cuticle present

(d) Chlorenchyma present

Identify the category of the plant and its part.

(1) Monocot stem (2) Monocot Leaf

(3) Dicot Root (4) Dicot Leaf

145. Which of the following is present on stroma lamellae and has reaction center P_{700}?

(1) PS II

(2) Cyt b_6f complex

(3) PS I

(4) ATP synthase

146. A bioactive molecule __________ is used as an immunosuppressive agent in organ transplant patients produced by a fungus *Trichoderma polysporum*.

Select the **correct** option to fill in the blank.

(1) Statins

(2) Amylase

(3) Cyclosporin A

(4) Streptokinase A

147. The pioneer community in hydrarch succession is

(1) Phytoplanktons (2) Bryophytes

(3) Lichens (4) Ferns

148. Egg apparatus of embryo sac has egg cell and synergids. The ploidy level and number of respective cells are

	Egg Cell		Synergids	
	Ploidy	Number	Ploidy	Number
(1)	n	1	n	1
(2)	2n	1	2n	2
(3)	n	1	n	2
(4)	2n	2	2n	1

149. The feature related to family Solanaceae is

(1) Trimerous flowers

(2) Epipetalous condition

(3) Persistent tepals

(4) Inferior ovary

150. Ribosomes

(1) Are single membrane bound

(2) Are not found within the cell organelle

(3) Are found in cytoplasm of prokaryotes

(4) Were discovered by Camillo Golgi

ZOOLOGY

SECTION-A

151. In a standard ECG, QRS complex represents

(1) Depolarisation of ventricles

(2) Repolarisation of ventricles

(3) End of ventricular systole

(4) Depolarisation of auricles

152. Match column I and column II and choose the **correct** option

	Column-I		Column-II
a.	hPL	(i)	Androgens
b.	Paired male sex accessory gland	(ii)	Cowper's gland
c.	Secondary oocyte	(iii)	Placental hormone
d.	Interstitial cells	(iv)	Forms acellular layer

Choose the **correct** option

(1) a - iii, b- ii, c- i, d- iv

(2) a - ii, b- i, c- iv, d- iii

(3) a - iii, b- ii, c- iv, d- i

(4) a - i, b- ii, c- iii, d- iv

153. IVF followed by ET in test tube baby programme includes techniques like

(1) GIFT and AI

(2) IUT and ZIFT

(3) AI and IUI

(4) ICSI and GIFT

154. Choose a combination of hormones which at high concentration facilitates ovulation.

(1) Estrogen, Progesterone, LH

(2) LH, FSH, Progesterone

(3) FSH, Estrogen, PRL

(4) Estrogen, LH, FSH

155. Select the **correct** statement.
(1) Muscularis is the innermost layer of digestive tract
(2) Ilium is a part of alimentary canal
(3) Vermiform appendix is tubular projection from rectum
(4) Ileum opens into large intestine

156. An example of unicellular gland is
(1) Salivary gland (2) Brunner's gland
(3) Goblet cell (4) Gastric gland

157. S.L. Miller created electric discharge in a closed flask and observed amino acids formation by mixing which of the following components at 800°C?
(1) H_2, O_2, CH_4, H_2O (2) CH_3, CO_2, H_2, H_2O
(3) CH_4, H_2, NH_3, H_2O (4) CH_3, H_2, SO_2, H_2O

158. The infective stage of malaria causing pathogen in humans and its site of storage in mosquito respectively are
(1) Trophozoite, hepatocytes
(2) Sporozoites, salivary glands
(3) Sporozoites, RBCs
(4) Gametocytes, hepatocytes

159. Linking of two fragments of DNA became possible with the enzyme
(1) Restriction endonuclease
(2) DNA ligase
(3) Alkaline phosphatase
(4) Catalase

160. Choose the **incorrect** statement w.r.t. nervous system of cockroach.
(1) Consist of series of fused segmentally arranged ganglia joined by connectives on the ventral side
(2) Three ganglia in the head and six in the thorax
(3) System is spread throughout the body
(4) Head holds a bit of a nervous system while the rest is situated along the belly part

161. The *ori* sequence controls
(1) Growth of recombinants
(2) Action of restriction endonucleases
(3) Copy number of the linked DNA
(4) Linking of antibiotic resistance gene in the plasmid

162. Match column I and column II and choose the correct option.

	Column-I		Column-II
a.	*cryIAc*	(i)	*A. tumefaciens*
b.	*Taq* polymerase	(ii)	*Thermus aquaticus*
c.	Native plasmid of *S. typhimurium*	(iii)	Specific Bt toxin gene
d.	Natural genetic engineer for plants	(iv)	First recombinant DNA

(1) a - iii, b- iv, c- ii, d- i
(2) a - iii, b- ii, c- iv, d- i
(3) a - i, b- ii, c- iii, d- iv
(4) a - ii, b- iii, c- iv, d- i

163. Match the following columns and choose the **correct** option.

	Column-I		Column-II
a.	Bacterial pathogen	(i)	Malaria
b.	Protozoan pathogen	(ii)	Elephantiasis
c.	Helminthic pathogen	(iii)	Pneumonia
d.	*Streptococcus* sp.	(iv)	Enteric fever

(1) a - iii, b- i, c- ii, d- iv
(2) a - iii, b- ii, c- i, d- iv
(3) a - ii, b- iv, c- i, d- iii
(4) a - iv, b- i, c- ii, d- iii

164. Which among the following induces the completion of the meiotic division in the secondary oocyte?
(1) Mere contact of sperm with zona pellucida
(2) Entry of sperm into cytoplasm of ovum
(3) Capacitation of sperms
(4) Release of first polar body

165. Choose the **correct** statement w.r.t. Ethidium bromide.
(1) It is used to separate DNA fragments in gel electrophoresis
(2) Aids in visualization of DNA fragments under UV light
(3) Enable the detection of RNA fragments under infrared radiation
(4) Produces bright blue light under UV radiation

166. A triploblastic acoelomate among the following is
(1) *Ascaris* (2) *Aurelia*
(3) *Fasciola* (4) *Pheretima*

167. Choose the **incorrect** match w.r.t. presence of bonds.
(1) Chitin – Glycosidic bonds
(2) Insulin – Peptide bonds
(3) Cholesterol – Phosphodiester bonds
(4) Cellulose – Glycosidic bonds

168. **Odd one** w.r.t. events happening during inspiration is
(1) Contraction of external inter-costal muscles
(2) Contraction of diaphragm
(3) Decrease in intra-pulmonary pressure
(4) Decrease in volume of thoracic chamber

169. Match the following column I and column II and choose the **correct** option.

	Column-I		**Column-II**
a.	Adaptive immune response	(i)	Neutrophils
b.	Associated with allergic reactions	(ii)	Lymphocytes
c.	Secrete serotonin and heparin	(iii)	Eosinophils
d.	Most abundant leucocyte	(iv)	Basophils

(1) a - ii, b- iii, c- iv, d- i (2) a - ii, b- iv, c- iii, d- i
(3) a - ii, b- iii, c- i, d- iv (4) a - i, b- ii, c- iv, d- iii

170. Which of the following is not an example of organs showing convergent evolution?
(1) Eye of *Octopus* and of mammals
(2) Sweet potato and potato
(3) Wings of butterfly and of birds
(4) Vertebrate hearts or brains

171. Choose the **incorrect** match
(1) PCT of nephron – Brush border cuboidal epithelium
(2) Ducts of salivary glands – Compound epithelium
(3) Eustachian tube – Simple squamous epithelium
(4) Lining of intestine – Columnar epithelium

172. Match the following column I and column II and choose the **correct** option.

	Column-I		**Column-II**
a.	Placoid scales	(i)	*Trygon*
b.	Operculated gills	(ii)	*Pristis*
c.	Ectoparasites on some fishes	(iii)	*Exocoetus*
d.	Sting ray	(iv)	*Petromyzon*

(1) a - ii, b- iii, c- iv, d- i (2) a - iii, b- iv, c- i, d- ii
(3) a - iv, b- ii, c- iii, d- i (4) a - i, b- iv, c- iii, d- ii

173. Collagen is the most abundant
(1) Protein in whole of the biosphere
(2) Protein in animal world
(3) Polysaccharide in animal world
(4) Lipid in whole of the biosphere

174. Select a palindromic sequence
(1) 5' GAATTC3'
3' CTTAAG5'
(2) 5' GAATTC3'
5' CTTAAG3'
(3) 5' GGTTTG3'
3' CCAAAC5'
(4) 5' TTGCCT3'
3' AACGGA5'

175. Which of the following is not an example of evolution by anthropogenic action?
(1) Industrial melanism
(2) Pesticide resistant varieties
(3) Herbicide resistant varieties
(4) Australian marsupials within Australian island

176. Match the following column I and column II and choose the **correct** option.

	Column-I		**Column-II**
a.	Shoulder joint	(i)	Scapula
b.	Triangular flat bone	(ii)	Articulates with acromion
c.	11 th and 12th pair of ribs	(iii)	Floating ribs
d.	Collar bone	(iv)	Glenoid cavity articulates with humerus

(1) a - ii, b- iii, c- i, d- iv (2) a - iv, b- i, c- iii, d- ii
(3) a - ii, b- i, c- iii, d- iv (4) a - iv, b- ii, c- iii, d- i

177. Choose the bacterial STI.
(1) Malaria (2) Cancer
(3) Trichomoniasis (4) Gonorrhoea

178. Restriction endonucleases
(1) Cleave only ssDNA at the terminal end
(2) Joins two DNA fragments by hydrogen bonds
(3) Cut the DNA strands at palindromic sites
(4) Always produce sticky ends

179. Select the factor which does not favour the formation of oxyhaemoglobin.
(1) Low pCO_2
(2) High pO_2
(3) Low temperature
(4) High H^+ concentration

180. A salt retaining hormone responsible for reabsorption of Na^+ and water from renal tubules is
(1) ANF (2) Renin
(3) ADH (4) Aldosterone

181. Most Bt toxins are
(1) Insect group specific
(2) Specific for fungal species
(3) Nematode group specific
(4) Specific for protozoans

182. '*Hisardale*' is a result of cross breed between
(1) Bikaneri rams and Marino rams
(2) Bikaneri ewes and Marino rams
(3) Bikaneri ewes and Marino ewes
(4) Bikaneri rams and Marino ewes

183. Read the following statements.
(a) Three subphyla of phylum Chordata are Tunicata, Cephalochordata and Vertebrata.
(b) In *Ascidia*, notochord is present only in larval tail.
(c) Notochord persists throughout life in all vertebrates.
(d) Nerve cord is ventral, solid and double in chordates.

Choose the option with only **incorrect** statements.
(1) (a) and (c) are correct
(2) (a) and (d) are correct
(3) (b) and (c) are incorrect
(4) (c) and (d) are incorrect

184. Select the **incorrect** match
(1) Glucagon – Hyperglycemic hormone
(2) Insulin – Hypoglycemic hormone
(3) Cortisol – Emergency hormone
(4) TCT – Hypocalcemic hormone

185. Match column I and column II.

	Column-I		Column-II
a.	Economically harmful gregarious arthropod	(i)	*Buthus*
b.	Radially symmetrical echinoderm	(ii)	*Ophiura*
c.	Chitinous exoskeleton and book lungs	(iii)	*Locusta*
d.	Ciliated comb plates	(iv)	*Pleurobrachia*

Choose the **correct** option
(1) a - iii, b- ii, c- i, d- iv
(2) a - i, b- ii, c- iii, d- iv
(3) a - ii, b- i, c- iii, d- iv
(4) a - iv, b- i, c- iii, d- ii

SECTION-B

186. An intestinal proteolytic enzyme that activates proteolytic biocatalyst of pancreatic juice is
(1) Carboxypeptidase (2) Enterokinase
(3) Pepsin (4) Rennin

187. Match the following column I and column II and choose the **correct** option.

	Column-I		Column-II
a.	Post transcriptional gene silencing	(i)	PCR
b.	DNA amplification	(ii)	Corrective therapy for hereditary diseases
c.	Gene therapy	(iii)	Insecticidal protein
d.	Bt toxin gene	(iv)	RNAi

(1) a - iv, b- iii, c- i, d- ii
(2) a - iv, b- ii, c- i, d- iii
(3) a - ii, b- iv, c- iii, d- i
(4) a - iv, b- i, c- ii, d- iii

188. Select the **incorrect** match

(1) Tyrosine – Aromatic amino acid

(2) Lysine – Basic amino acid

(3) Glutamic acid – Acidic amino acid

(4) Valine – Alcoholic amino acid

189. Glycosuria and ketonuria in urine helps in clinical diagnosis of

(1) Vasopressin deficiency mediated disorder

(2) Disease caused by ADH deficiency

(3) Disease caused by insulin deficiency

(4) Glucagon deficiency mediated disorder

190. Match the following column I and column II and choose the **correct** option.

	Column-I		Column-II
a.	Auditory receptors	(i)	Eustachian tube
b.	Pressure equalizer between middle ear and outer ear	(ii)	Organ of Corti
c.	Vestibular apparatus	(iii)	Increase efficiency of sound wave transmission
d.	Ear ossicles	(iv)	Above coiled portion of labyrinth called cochlea

(1) a - i, b- iv, c- iii, d- ii

(2) a - ii, b- i, c- iv, d- iii

(3) a - i, b- ii, c- iv, d- iii

(4) a - ii, b- iv, c- iii, d- i

191. Choose a set of secondary metabolites

(1) Caffeine and glucose

(2) Strychnine and valine

(3) Cellulose and codeine

(4) Anthocyanin and chlorophyll

192. Functional and fully mature form of insulin has

(1) Two polypeptide chains A and B mainly linked by glycosidic bonds

(2) An extra peptide chain called C chain

(3) Two short polypeptides linked together by disulphide bridges

(4) One interchain and two intrachain disulfide bridges

193. Read the following statements and choose the incorrect statement.

(1) Active immunity is a slow immune response but pathogen specific

(2) IgG and IgA imparts passive immunity to foetus

(3) Anti-tetanus serum is an example of passive immunity

(4) Active immunity is exemplified by a condition when a snake bites and host's body naturally produces antibodies against snake venom

194. Select the **incorrect** match

(1) Alfred Wallace – Malay Archipelago

(2) Oparin and Haldane – Life could have come from pre-existing non-living organic molecules

(3) Karl Ernst von Baer – Biogeographical evidences of evolution

(4) Charles Darwin – Natural selection as mechanism of evolution

195. Match the following column I and column II and choose the **correct** option.

	Column-I		Column-II
a.	Abrin	(i)	Succinate dehydrogenase inhibitor
b.	Oxaloacetate	(ii)	Polymer of N-acetylglucosamine
c.	Exoskeleton of arthropods	(iii)	Posses peptide bonds
d.	GLUT-4	(iv)	Toxin, secondary metabolite

(1) a - iv, b- iii, c- ii, d- i

(2) a - iii, b- ii, c- i, d- iv

(3) a - iv, b- i, c- ii, d- iii

(4) a - i, b- iii, c- iv, d- ii

196. Match the following

	Column-I		Column-II
a.	Exopthalmic goitre	(i)	Adrenal gland
b.	Insulin deficiency	(ii)	Thyroid gland
c.	Aldosteronism	(iii)	Pituitary gland
d.	Gigantism	(iv)	Pancreas

Choose the correct option from the following

(1) a -ii, b- iv, c- i, d- iii

(2) a - i, b- ii, c- iii, d- iv

(3) a - iv, b- ii, c- i, d- iii

(4) a - iv, b- iii, c- ii, d- i

197. Select the option representing structures included in forebrain only.

(1) Corpus callosum, cerebellum, hypothalamus

(2) Thalamus, hypothalamus, corpora quadrigemina

(3) Cerebral aqueduct, cerebrum, association areas

(4) Cerebrum, thalamus, hypothalamus

198. Which hominid arose during ice age between 75,000 – 10,000 years ago?

(1) *Homo erectus*

(2) *Homo habilis*

(3) Modern *Homo sapiens*

(4) Neanderthal man

199. Among the given sets of contraceptive methods, which set shares similar mode of action?

(1) Saheli, Diaphragm

(2) Multiload 375 and Progestasert

(3) Lippes loop, Vasectomy

(4) CuT and Multiload 375

200. The human protein-enriched milk is produced by first transgenic cow named

(1) Tracy (2) Rosie

(3) Dolly (4) ANDi

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